

YaP Ability VFX — quality upgrade (V1-V4)

Goal: Bring spell/ability visuals to top-MMO-plugin quality without abandoning Folia safety or the YAML ability packs.

Depends on: M6-M7 ability engine (`yap-abilities.jar` , `docs/mmo/MMO_ABILITIES.md`)

Status:  **V1-V4 complete** (2026-09-04)

Phase map

Phase	Name	Status	Ships
V1	Engine foundations	 Done	<code>yap-abilities-api</code> + <code>yap-abilities.jar</code> , <code>showcase_vfx.yml</code>
V2	Template / generator overhaul	 Done	<code>scripts/generate-ability-pack.py</code> + regenerated bulk YAML
V3	Hero ability art pass	 Done	<code>showcase_heroes.yml</code> (12 signature casts)
V4	Polish & crossplay	 Done	Status VFX, Bedrock names, unique hero icons, Folia soak gates

```
flowchart LR
  V1[V1 Engine] --> V2[V2 Templates]
  V1 --> V3[V3 Heroes]
  V2 --> V3
  V2 --> V4[V4 Polish]
  V3 --> V4
```

Rules (every phase)

- World/entity work via `YapSched` only
- Respect `folia-safe.*` caps in `YaPAbilities/config.yml`
- Prefer declarative YAML over per-spell Java
- Do **not** hand-edit bulk generated packs at scale — change the generator (V2)
- Hand-author only hero/showcase abilities

V1 — Engine foundations

Status:  Done

Feature	YAML	Runtime
Non-blocking timed steps	at: N	EffectRunner
Shake	type: shake	ImpactFx
Arc paths	projectile.path: arc	ProjectileTracker
Trail styles	trail.style + falloff	motion / ribbon / burst
Impact weight	impact-shake	staged nova + orb + shake
New shapes	cone pillar orb shockwave	VfxEmitter

Showcase: showcase_vfx.yml · Tests: AbilityPackLoaderVfxTest

V2 — Template / generator overhaul

Status:  Done

- Script: scripts/generate-ability-pack.py
- Kits: fire / water / wind / earth / arcane / curse / prayer / melee / ranged / utility
- Regenerated **227** abilities; gameplay stats preserved
- Skips showcase_*.yaml

```
python3 scripts/generate-ability-pack.py
python3 scripts/generate-ability-pack.py --dry-run
```

V3 — Hero ability art pass

Status:  Done · Pack: abilities/showcase_heroes.yaml

Id	School
inferno_meteor	Fire
tidal_collapse	Water
cyclone_rift	Wind
tectonic_spike	Earth
void_lance	Arcane
hex_bloom	Curse

Id	School
solar_aegis	Prayer
blade_tempest	Melee
skyfall_volley	Ranged
phase_mirror	Utility
dragonfire_cascade	Fire ultimate
judgment_smite	Melee / prayer

V4 — Polish & crossplay

Status:  Done

Item	Evidence
Status-effect ambient VFX	effects/common.yml shaped/colored ticks
Folia projectile headroom + author notes	config.yml (max-projectiles-per-player: 14 , global 112)
Bedrock cast labels with ability name	AnimationSync + EffectRunner
Unique hero icons (78020-78031)	scripts/generate-hero-ability-icons.py → pack textures/models
Folia VFX soak gate (offline)	scripts/content/ability-vfx-soak-gate.py + AbilityVfxSoakGateTest

```
python3 scripts/generate-hero-ability-icons.py
python3 scripts/content/ability-vfx-soak-gate.py
gradle :abilities-plugin:test --tests 'com.yapcore.abilities.load.*'
```

Latest soak gate sample: **249** abilities · **486** timed at: steps · **283** shakes · **107** arcs · **156** motion/ribbon trails · shapes include cone/pillar/orb/shockwave · **PASS**

Optional live soak (when the game server is up): cast inferno_meteor , void_lance , dragonfire_cascade under multiplayer spam and confirm caps hold.

V1 YAML cheat sheet

```
cast:
  - type: animation
    style: cast
  - type: vfx
    particle: FLAME
    shape: helix
  - type: vfx
    at: 4
    particle: DUST
    shape: ring
    color: 255,80,20
  - type: shake
    power: 0.08
    pulses: 2
```

```
projectile:
  entity: SNOWBALL
  path: arc
  arc-height: 2.2
  hide: true
  impact-shake: true
  trail:
    particle: FLAME
    count: 6
    interval: 1
    style: motion
    falloff: 0.55
```

```
on-hit:
  - type: damage
    style: magic
    max-hit: 10
  - type: vfx
    particle: LAVA
    shape: shockwave
    count: 20
    radius: 2.0
  - type: shake
    power: 0.2
    pulses: 3
```

Shapes: burst · ring · helix · beam · nova · cone · pillar · orb · shockwave